

10/9/23

Mathematics

(1)

1 3

2 No solⁿ

3 10 units

4 Proportional

5 $-\frac{1}{2}$ 6 $x^3 y^3$ 7 Sum of zeroes = $\frac{1}{2}$ Their product

$$\alpha + \beta = \frac{1}{2} \alpha \beta$$

$$\frac{-b}{a} = \frac{1}{2} \left(\frac{c}{a} \right)$$

$$\frac{1}{2} = \frac{k-4}{8}$$

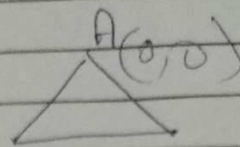
$$8 = 2k - 8$$

$$2k = 16$$

$$k = 8$$

8

$AB^2 = AC^2$ (because it's an equilateral Δ^{ic})



$$(3-0)^2 + (\sqrt{3}-0)^2 = (3-0)^2 + (x-0)^2$$

B (3, $\sqrt{3}$) C (3, x)

$$\Rightarrow (\sqrt{3}-0)^2 = (x-0)^2$$

$$x = \sqrt{3} //$$

9 $2px + 3y = 7$

$$2x + y = 6$$

$$\frac{2p}{2} \neq \frac{3}{1} \text{ (Given)}$$

$$2p \neq 6$$

$$p \neq 3 //$$

16 $PD \parallel BE$

$$\frac{AP}{PB} = \frac{AD}{DL} \text{ (B.P.T.)} \rightarrow (1)$$

$$\frac{AD}{DC} = \frac{CE}{BE} \text{ (Given)} \rightarrow (2)$$

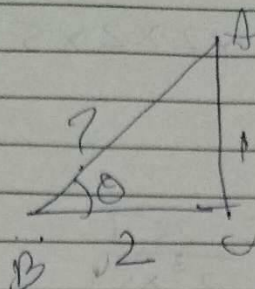
$$\frac{AP}{PB} = \frac{CE}{BE} \text{ (From (1) \& (2)}$$

by Converse of BPT
 $DE \parallel BAC$

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$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$



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$$\begin{aligned} AB^2 &= AC^2 + BC^2 \\ &= 1 + 4 \\ &= 5 \end{aligned}$$

$$AB = \sqrt{5}$$

$$\sin \theta = \frac{\text{opp}}{\text{hypotenuse}}$$

$$= \frac{1}{\sqrt{5}}$$

~~$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$~~

$$\cos \theta = \frac{\text{adj}}{\text{hypotenuse}}$$

$$= \frac{2}{\sqrt{5}}$$

$$\frac{1}{2} \sin \theta \cos \theta = \frac{1}{5}$$

$$\frac{1}{2} \times \frac{1}{\sqrt{5}} \times \frac{2}{\sqrt{5}} = \frac{1}{5}$$

$$\frac{1}{5} = \frac{1}{5} \text{ Hence proved}$$

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$$12 \quad 5000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 5$$

$$= \sqrt[4]{2^3 \times 5^4}$$

$$\frac{257}{5000} = \frac{257}{2^3 \times 5^4}$$

$$\Rightarrow \frac{257}{2^3 \times 5^4} \times \frac{2}{2}$$

$$\frac{514}{(10)^4}$$

$$= 0.0514$$

$$\begin{array}{r} 1 \\ 257 \times 2 \\ \hline 514 \end{array}$$

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$$\sqrt{3} = \frac{p}{q}$$

Let us first assume that $\sqrt{3}$ is rational and is in the form p/q and $q \neq 0$ (p, q are coprime integers)

Square on both sides

$$3 = \frac{p^2}{q^2}$$

$$3q^2 = p^2$$

$$\Rightarrow 3 \text{ is a factor of } p^2$$

$\therefore 3$ is a factor of p also

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$$p = 3k$$

Square on Both sides

$$p^2 = 9k^2$$

$$3q^2 = 9k^2$$

$$q^2 = 3k^2$$

$\Rightarrow 3$ is a factor of q^2

$\therefore 3$ is a factor of q also

3 is now the factor of both p and q

This contradicts our statement that p and q are co-prime integers

$\therefore \sqrt{3}$ is an irrational no.

16. Given $\triangle ABE \cong \triangle ACD$

$$AB = AC \text{ (C.P.C.T)} \rightarrow (1)$$

$$BE = CD \text{ (C.P.C.T)} \rightarrow (2)$$

$$\triangle ABE \cong \triangle ACD \rightarrow (3)$$

Dividing (3) by (1)

$$\frac{AD}{AB} = \frac{AE}{AC}$$

In $\triangle ADE$ & $\triangle ABC$

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$\angle A = \angle A$ (common)

$\frac{AD}{AB} = \frac{AE}{AC}$ (proved)

By SAS criterion

$\triangle ADE \sim \triangle ABC$

Hence proved

$$\frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta}{\cos^2 \theta} - 1} + \frac{1}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

$$\frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}} + \frac{1}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

$$\frac{\frac{\sin^2 \theta}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \cos^2 \theta}} + \frac{1}{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \cos^2 \theta}}$$

$$\frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\sin^2 \theta \cos^2 \theta}{\sin^2 \theta (\sin^2 \theta - \cos^2 \theta)} \quad (4)$$

$$\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta} = \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

= R.H.S

Hence proved

Hence proved

19 let the fraction be $\frac{x}{y}$

$$\frac{2x}{y+1} = 1 \quad (\text{Given})$$

$$2x = y+1$$

$$\frac{x+4}{2y} = \frac{1}{2} \Rightarrow x+4 = y$$

$$2x - y - 1 = 0$$

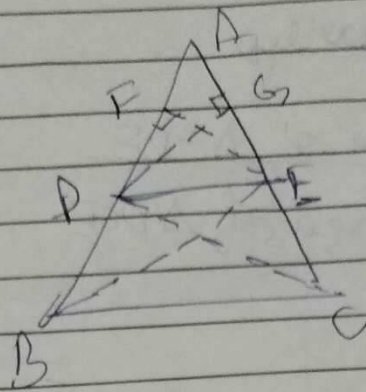
$$x - y + 4 = 0$$

$$x = -5$$

$$x+5+4=0 \quad -5-y+4=0$$

$$y = -9$$

$\therefore$ The fraction is $\frac{5}{9}$



20 BPT states that, if a line parallel to a side of a triangle intersects the other 2 sides then the line divides it into equal proportion

ABC be a Δ^e

$BC \parallel DE$

To prove $\frac{AD}{DB} = \frac{AE}{EC}$

Join BE, CD

Construction: $EF \perp AB$, $DG \perp CA$

$\therefore$ Area of $\triangle ADE = \frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times AD \times EF$$

Area of $\triangle DBE = \frac{1}{2} \times AE \times DG$

$$\frac{\text{Area of } \triangle ADE}{\text{Area of } \triangle DBE} = \frac{AD}{DB} \Rightarrow \text{①}$$

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Similarly

$$\frac{\text{area of } \triangle ADE}{\text{area of } \triangle DCE} = \frac{\frac{1}{2} \times AE \times DG}{\frac{1}{2} \times EC \times DG}$$

$$= \frac{AE}{EC}$$

$\triangle DBE$ and $\triangle DCE$ have the same base b/w same parallels.

$$\therefore \frac{AD}{AB} = \frac{AE}{AC} \quad \therefore \frac{AD}{DB} = \frac{AE}{EC}$$

$$21 \quad \frac{1}{2} + 2 \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{\sqrt{3}}{1} \right)^2$$

$$\frac{1}{2} \left(\frac{1}{1} \right) + \left(\frac{\sqrt{3}}{2} \right)^2 + 1^2$$

$$\frac{1}{2} + \frac{2}{2} + 3$$

$$\frac{1}{2} + \frac{3}{2} + 1$$

$$\frac{1}{2} + \frac{2}{2} + 3$$

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$$\frac{1}{2} + \frac{3}{4} + 1$$

$$\frac{3}{2} + 3$$

$$\frac{1}{2} + \frac{3}{4} + 1$$

$$\frac{3+6}{2}$$

$$\Rightarrow \frac{9}{2} \times \frac{42}{9}$$

$$\frac{2+3+4}{4}$$

$$\therefore \sin 30 + 2\cos^2 45 + \tan^2 60 = 2$$

$$\frac{1}{2} (\cot^2 45 + \cos^2 30 + \tan^2 45)$$

covered by

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$$\text{Time } A = \frac{1980}{330} = 6 \text{ min}$$

(One round)

$$\text{Time covered by } B = \frac{1980}{198} = 10 \text{ min}$$

(one round)

$$\text{Time } C = \frac{1980}{220} = 9 \text{ min}$$

(One round)

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Father's age be $2x$
 Son's age be y

$$(2y + x = 70) \times 2$$

$$2x + y = 95$$

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$$4y + 2x = 140$$

$$y + 2x = 95$$

$$\Rightarrow 3y = 45$$

$$y = 15$$

$$15 + 2x = 95$$

$$2x = 80$$

$$x = 40$$

Father's age 40

Son's age 15

25 $(6, 4)$ and $(1, -7)$ be divided by the point on x -axis $(x, 0)$

let us assume the ratio to be $k:1$

$$\frac{-7k + 4}{k + 1} = 0 \quad (\text{Section formula})$$

$$-7k + 4 = 0$$

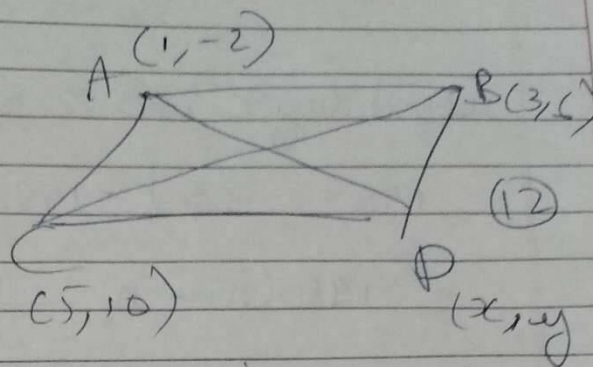
$$k = \frac{4}{7}$$

$$m:n \Rightarrow \frac{4}{7}:1 \Rightarrow 4:7$$

$$m + n = 4 + 7 = 11$$

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W.K.T
diagonals
of a $\square$ bisect
each other



So the mid point of AC is same as
mid point of BD

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{3 + x}{2}, \frac{6 + y}{2} \right)$$

$$\left(\frac{1 + 5}{2}, \frac{-2 + 10}{2} \right) = \left(\frac{3 + x}{2}, \frac{6 + y}{2} \right)$$

$$\frac{6}{2}, \frac{8}{2} = \frac{3 + x}{2}, \frac{6 + y}{2}$$

$$3 + x = 6; 6 + y = 8$$

$$x = 3; y = 2$$

So ~~the~~ Fourth vertex's coordinates
are (3, 2)

23rd question continuation

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A, B meet after time t and completed n_1, n_2
 n_3 rounds respectively

$$330t = x + 1980n_1 \rightarrow (1)$$

$$198t = x + 1980n_2 \rightarrow (2)$$

$$198t = x + 1980n_3 \rightarrow (3)$$

PROCEED

substituting x into (2) & (3)

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$$198t = 330t - 1980n_1 + 1980n_2$$

$$1980(n_1 - n_2) = 132t$$

$$t = 15(n_1 - n_2)$$

$$\text{Also, } 220t = 330t - 1980n_1 + 1980n_3$$

$$1980(n_1 - n_3) = 110t$$

$$t = 18(n_1 - n_3)$$

Since t is same and $n_1 - n_3$ and $n_1 - n_2$ can be anywhere no.

LCM of 15 and 18
is 90

A, B, C will meet after 90 mins

28.1)

$$D \text{ of PR} = \frac{8-2}{2} / \frac{3-5}{2}$$

$$= \frac{6}{2} / \frac{-2}{2}$$

$$= 3, -1$$

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Ex 4. Relative speed $x - y$ (same direction)

$$\text{Dist} = (x - y) \times 5$$

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$$x - y = 20$$

$x + y$ (opposite direction)

$$x + y = 100$$

$$x - y = 20$$

$$x + y = 100$$

$$2x = 120$$

$$x = 60 \text{ km/h}$$

$$y = 40 \text{ km/h}$$

30 $\triangle ABD \sim \triangle PGD$ (Corresponding sides are parallel)

$$\frac{AC}{BD} = \frac{AP}{GP} \Rightarrow \frac{1}{x} = \frac{BP}{z \times BP} \rightarrow \text{①}$$

$\triangle CPB \sim \triangle GDB$ (Corresponding sides are parallel)

$$\frac{AC}{BD} = \frac{CP}{GD} \Rightarrow \frac{1}{y} = \frac{BP}{z \times BP} \rightarrow \text{②}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{BP + GP}{z \times BP} = \frac{1}{z}$$

// Hence proved
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$$x + B = -\frac{3}{2}$$

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$$x B = \frac{7}{2}$$

$$\frac{1}{x^2} + \frac{1}{B^2} \Rightarrow \frac{x^2 + B^2}{x^2 B^2}$$

$$= \left(\frac{-3}{2} \right)^2 - \left(2 \times \frac{7}{2} \right)^2$$

$$\left(\frac{7}{2} \right)^2$$

$$= \frac{9 - 7}{4}$$

$$49$$

$$= -\frac{18}{49}$$

$$\text{Also } \frac{1}{(x B)^2} = \frac{4}{49}$$

$$x^2 + \left(\frac{18}{49} \right) x + \frac{4}{49}$$

$$49x^2 + 18x + 4 = 0 //$$

18 $\frac{3}{5} = \frac{-a-1}{1-2a}$

$$-5(a+1) = 3(1-2a)$$

$$a = 8$$

$$\frac{3}{5} = \frac{2b-1}{3b}$$

$$9b = 10b - 5$$

$$b = 5$$